Meme Coin Analysis Dashboard

Data Engineering Project Proposal



much data.
such pipeline.

IDSC501

Team members: TBD

Doge illustration: Icons8

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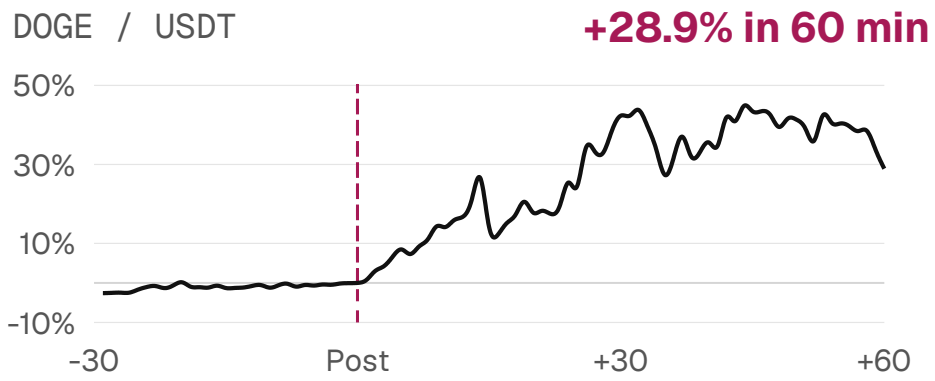
Why meme coins?

When they post

Elon Musk

“Doge”

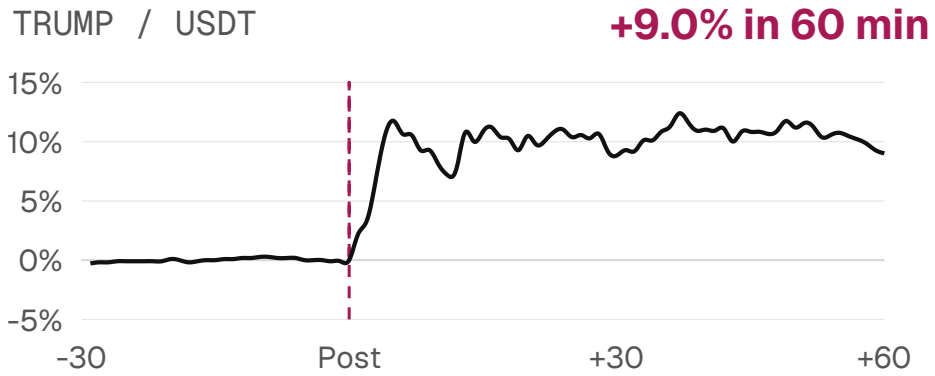
X (Twitter)
04 Feb 2021



Donald Trump

“I LOVE $TRUMP”

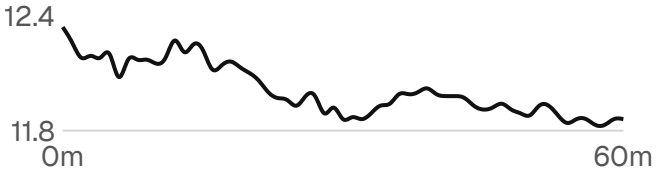
Truth Social
23 Mar 2025



When we buy…



TRUMP after the spike



16:09–17:09 UTC, 23 Mar 2025. USDT.
Meme caption, not our trade.

Can prices + posts predict the next move?

Binance 1m closes. X: minutes from post. Selected events, not causal proof. Art: UniqueDesignOK.

Data collection

14,773 price rows 10 coins · social collection planned

Source	Collection method	Output
CoinMarketCap Historical Data	**Web crawling**	Daily price CSV
X via FxTwitter	**Third-party API**	Posts + reactions in JSON
Reddit via Arctic Shift	**Archive API / files**	Posts + comments

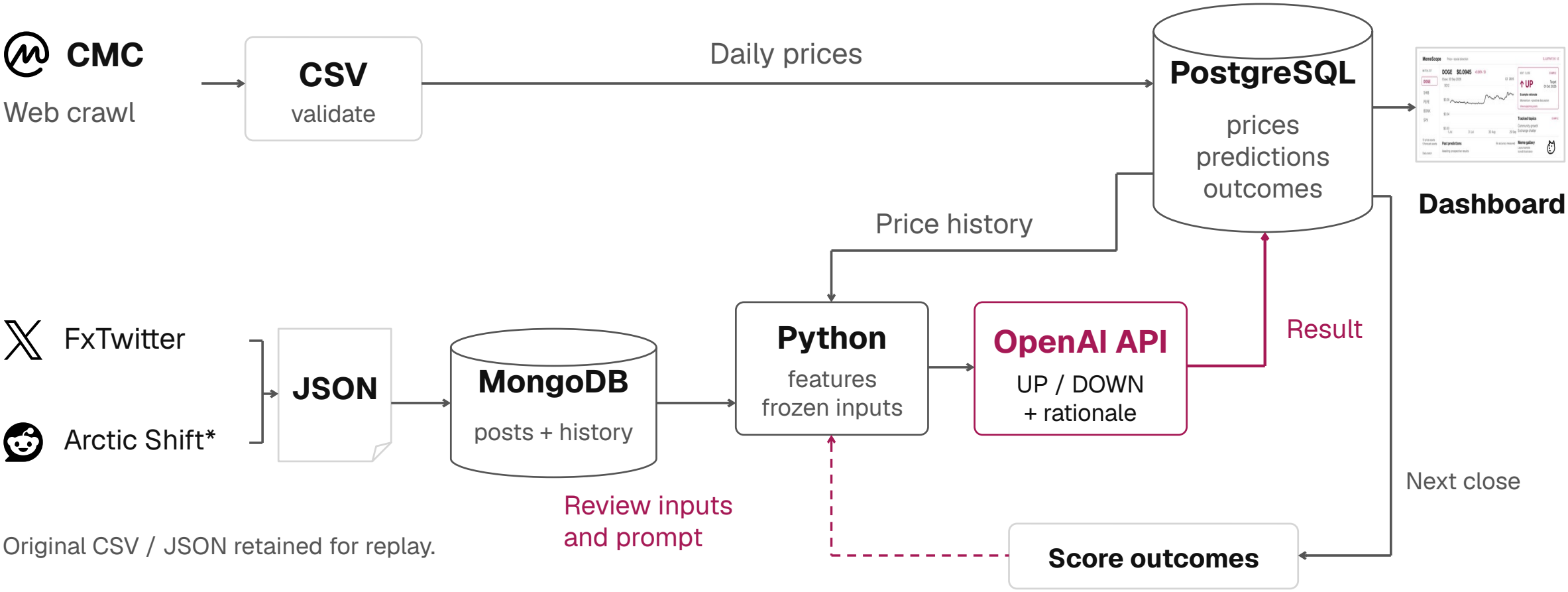
Daily social batches, with a seven-day overlap.

Project websites: crawl links and contract details to verify sources.

Reddit awaits access conditions. Price refresh needs new CSVs or restoration of the team crawler.

Data pipeline

Next UTC close: UP or DOWN



Dashed: team review for future versions. Post details read MongoDB. * Reddit: conditional access.

Data preprocessing

Only information available when the forecast is issued

Price features

31 daily closes
Returns and volatility
Volume changes

Social features

Activity by source
Sentiment + coverage
Selected post excerpts

Frozen snapshot

Known before cutoff
Stable IDs and hashes
Model + prompt version

UTC normalization · ID deduplication · verified coin matching

No future information

Later posts and updated reactions stay out of earlier inputs.

Freeze at 03:00 UTC. Publish by 03:15. Missing data or failed inference remains unavailable.

Why two databases?

PostgreSQL can also store JSON.

MongoDB

Social documents

Posts and observations
Platform-specific fields
Model input / output

PostgreSQL

Relational analytics

Prices and daily features
Predictions and outcomes
Keys, constraints and joins

Document model → ETL → relational model

Document storage and relational analysis, connected by a traceable ETL.

Trade-off: two databases, indexes, backups and consistency checks.

Prediction dashboard

MemeScope Price + social direction ILLUSTRATIVE UI

WATCHLIST

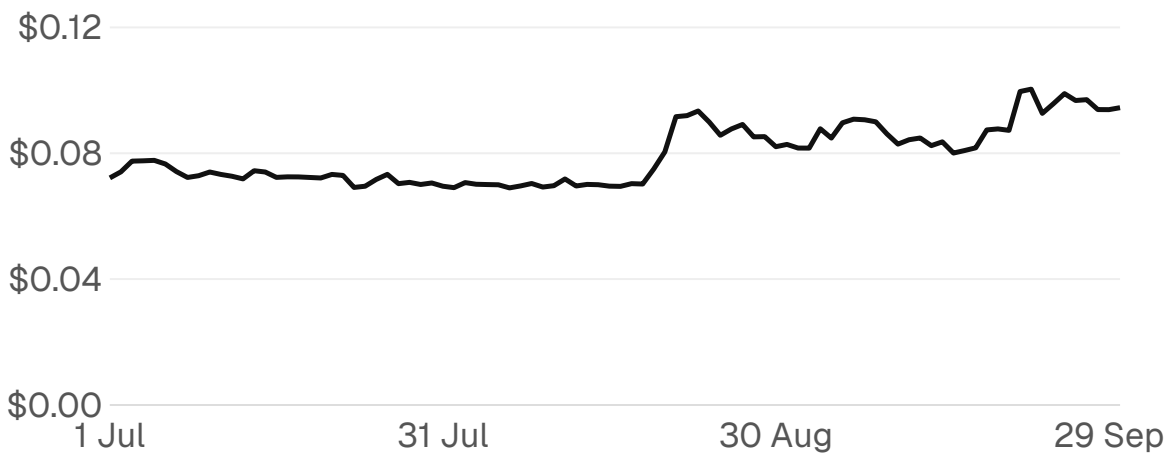
- **DOGE**
- SHIB
- PEPE
- BONK
- SPX

10 price assets
5 forecast assets

Daily batch

DOGE $0.0945 +0.66% 1D

Close: 30 Sep 2026 Q3 2026



Past predictions

No accuracy measured

Awaiting prospective results

NEXT CLOSE EXAMPLE

↑ UP

Target
01 Oct 2026

Example rationale

Momentum + positive discussion

View supporting posts

Tracked topics

EXAMPLE

Community growth
Exchange chatter

Meme gallery

Layout sample
Icons8 illustration



Supplied DOGE prices. Prediction and social text are examples. Image: Icons8.